



SIMATS
ENGINEERING



SIMATS
Saveetha Institute of Medical And Technical Sciences
(Declared as Deemed to be University under Section 3 of UGC Act 1956)

SAVEETHA SCHOOL OF ENGINEERING, SIMATS
Department of Computer Science and Engineering

Assignment Information

Particular	Details
Name:	Atmakuri Sai Mrudula
Register Number:	192372087
Department:	B.E.CSE (Artificial Intelligence)
Programme:	CSE
Course Code & Course Name	CSA1522 – Cloud Computing and Big Data Analytics
Academic Year / Batch	2026–27
Faculty Name	Dr P Rajaram
Assignment Title	Cloud-Based Placement Data Management and Big Data Analytics using JOHO
Date of Issue	31-08-2026
Date of Submission	01-09-2026
Maximum Marks	100
Course Outcome(s) – CO	CO1: Understand the cloud services with different service models. CO2: Demonstrate the deployment of Virtualization and build a data center using virtualization tools. CO3: Build methods to access cloud and various techniques to collaborate within cloud.
Bloom's Taxonomy Level	L3 – Apply and L4 – Analyze
SDG Mapping	SDG 4 – Quality Education SDG 8 – Decent Work and Economic Growth SDG 9 – Industry Innovation and Infrastructure SDG 11 – Sustainable Cities and Communities
Industry	Applies cloud services, virtualization and distributed big-data processing to a ZOHO-based placement workflow, supporting scalable access, collaboration and analytics while considering security, privacy and responsible data use.

Problem Statement:

University placement cells handle a large amount of information related to students, companies, job opportunities, applications, interviews, and final placement results. Managing this information using traditional methods such as spreadsheets, paper records, or separate computer systems can lead to data duplication, loss of information, difficulty in updating records, and unauthorized access. Students may also face difficulties in accessing current placement opportunities and application status, while placement staff require a centralized system to efficiently manage and share placement information.

Assessment Rubrics

Assessment Criterion / Marks	Excellent (4)	Good (3)	Satisfactory (2)	Needs Improvement (1)
CO1 – Cloud Service Models & Architecture 25 Marks	Correctly identifies the JOHO requirements, selects appropriate IaaS/PaaS/SaaS models, and presents a complete architecture with clear justification.	Selects appropriate service models and presents a suitable architecture with minor gaps in justification or detail.	Identifies basic service models and provides a simple architecture, but the justification or mapping to requirements is limited.	Service models are unsuitable or misunderstood; architecture is missing/unclear and justification is inadequate.
CO2 – Virtualization & VM Deployment 25 Marks	Creates and configures two VMs correctly using Oracle VM VirtualBox; records CPU, memory, storage and networking; demonstrates communication and clearly explains the hypervisor.	Creates and configures both VMs successfully with mostly correct resources and networking; communication works with minor issues.	Creates the VMs and demonstrates basic operation, but configuration, networking, testing or explanation is incomplete.	VM creation/configuration is incorrect or incomplete; communication is not demonstrated and supporting evidence is insufficient.
CO3 – Cloud Access & Collaboration 25 Marks	Demonstrates secure cloud access successfully, including authentication/access control, clear request-response evidence and a practical collaboration workflow.	Demonstrates working cloud access and collaboration with minor gaps in security, explanation or evidence.	Demonstrates basic cloud access, but collaboration, authentication or security evidence is limited.	Cloud access or collaboration is non-functional, unclear, insecure or not supported by evidence.
Analysis, Validation & Engineering Justification 15 Marks	Compares feasible alternatives using clear criteria, performs relevant tests, interprets results accurately and gives a well-supported final decision.	Provides a reasonable comparison and testing with generally correct interpretation and justification.	Provides limited comparison or testing; interpretation and justification are basic.	Little or no meaningful comparison, testing, interpretation or engineering justification is provided.
Security, Privacy, Ethics & SDG Relevance 10 Marks	Clearly addresses security, privacy, ethical handling of placement data and relevant SDGs with specific application to the JOHO scenario.	Addresses relevant security, privacy, ethical and SDG considerations with minor omissions.	Mentions general security, privacy, ethics or SDG points but gives limited connection to the problem.	Important security, privacy, ethical or SDG considerations are missing or inappropriate.

Deliverables

To receive full credit, each submission must include the following deliverables:

1. Pseudo code
2. Implementation & Results
- 3.** Github and GCR Upload

Assignment Problem / Challenge

The university placement cell faces difficulties in managing student, company, job, application, interview, and placement information using traditional methods. Maintaining these records manually or across separate systems can result in data duplication, information loss, delayed updates, and unauthorized access. Students may also find it difficult to access the latest placement opportunities and application status.

The challenge is to design and evaluate a simple cloud-based Placement Data Management and Access System using Zoho Creator. The system should provide a centralized platform where placement staff can manage student and company information, while students can securely access eligible job opportunities, submit applications, and track their placement status.

The assignment also requires demonstrating the IaaS, PaaS, and SaaS cloud service models. Oracle VM VirtualBox will be used to create and configure a virtual machine to demonstrate the IaaS concept, while Zoho Creator will serve as the PaaS platform for developing the placement application, managing forms, workflows, databases, and user access. The completed Zoho-based placement application will demonstrate the SaaS model by allowing authorized students and placement staff to access the system through the web. The solution should also demonstrate secure access, Web API usage, role-based permissions, cloud data sharing, and collaboration among users.

Problem Statement

The university placement cell currently faces challenges in managing a large volume of student and company placement information efficiently. Student details, company profiles, job opportunities, applications, interview schedules, and selection results may be stored using different systems or manual records, making the data difficult to maintain and access.

The major challenge is to design a reliable and secure cloud-based placement data management system that allows students and placement staff to access the required information based on their roles. The solution must ensure that sensitive student and placement data is protected from unauthorized access while allowing authorized users to collaborate and update information.

The assignment also requires demonstrating how IaaS, PaaS, and SaaS cloud service models can be applied to the placement system. A virtual machine must be created and configured using Oracle VM VirtualBox, while a cloud-hosted application or Web API must demonstrate secure access to placement data. The system should therefore provide centralized data management, controlled access, scalability, security, availability, and cloud-based collaboration for the university placement process.

Requirements and Constraints Functional

Requirements

The proposed cloud-based placement management system shall provide a centralized platform for managing university placement information. The system shall allow students to register and maintain their academic and placement profiles, while placement staff can manage student records, company details, job opportunities, applications, interview schedules, and placement results. Students shall be able to view suitable job opportunities, check eligibility, apply for companies, and track their application status. Placement staff shall be able to update application and selection status and generate placement-related reports. The system shall also provide role-based access so that students, placement staff, and administrators can access only the functions relevant to their responsibilities.

Technical Requirements

The system requires a computer with a stable Internet connection and a modern web browser for accessing the cloud application. Zoho Creator is used as the cloud-based development platform for creating forms, databases, workflows, reports, dashboards, and user access controls. VMware Workstation Pro is used to create and configure the Placement-Server and Placement-Client virtual machines. The virtual machines can use Ubuntu/Linux operating systems and can be configured with suitable CPU, RAM, storage, and network resources. A Host-only network is used to establish controlled communication between the two virtual machines for testing. The system may also use Web/API functionality for secure communication and data access.

Security & Privacy

Security is important because the system contains student academic details, contact information, resumes, application records, and placement results. The application shall use authenticated user access and role-based permissions. Students should only be able to view or modify their own information and should not be able to access other students' confidential records. Placement staff shall have additional privileges for managing placement information. Secure communication such as HTTPS should be used when accessing the cloud application. Sensitive information should not be unnecessarily exposed, and access to shared resources should be restricted to authorized users.

Reliability

The system should provide reliable access to placement information whenever required. Cloud-based storage through Zoho Creator reduces dependence on individual computers and provides centralized access to information. Proper data validation, controlled access, regular monitoring, and appropriate backup mechanisms can improve reliability. The virtual machines should also be configured correctly so that network communication and application testing can be performed consistently.

Constraints

The system is dependent on Internet connectivity and the availability of the Zoho Creator platform. The functionality of the application may also depend on the features and limitations of the selected Zoho Creator plan. The virtual machines have limited computing resources and are mainly intended for demonstration and testing rather than production deployment. The project focuses on basic placement management and does not implement advanced features such as artificial intelligence-based job recommendations or large-scale enterprise integration. User access is also limited according to the permissions defined by the placement cell.

Student Work

The student's work focused on the complete practical implementation of the proposed cloud-based placement management solution. This included developing the application using Zoho Creator, implementing placement data management and access features, configuring VMware Workstation Pro virtual machines, establishing network communication between the Placement-Server and Placement-Client using a Host-only network, testing the complete system, verifying the results, and preparing the required documentation and screenshots.

Problem Understanding and Formulation

What is the problem

University placement cells manage large quantities of student, company, job, application, interview, and selection information. When this information is maintained using manual records or separate files, it can become difficult to update, share, protect, and retrieve. Students may not receive timely information about available opportunities, while placement staff may face difficulties in maintaining accurate and centralized records.

Therefore, the problem is to develop a simple, secure, and cloud-based placement data management and access solution using Zoho Creator, supported by virtualization and networking demonstrations.

Expected Outcomes

The expected outcome is a working cloud-based placement management application that allows authorized students and placement staff to manage and access placement information efficiently. The project should demonstrate the practical concepts of IaaS, PaaS, SaaS, virtualization, networking, security, cloud collaboration, and Web/API access. The completed system should improve accessibility, organization, controlled data sharing, and placement information management.

Information/Data Available

The system can manage the following information:

- Student ID and name
- Department and academic details
- CGPA
- Skills
- Contact information
- Resume
- Company details
- Job role
- Salary/package
- Eligibility criteria
- Application details
- Interview schedule
- Shortlisting status
- Selection/rejection status
- Final placement information

Assumptions

The project assumes that students and placement staff have valid user accounts and access to the Internet. It is assumed that the placement cell provides accurate student and company information. It is also assumed that Zoho Creator is available for developing and hosting the application. The virtual machines are assumed to have sufficient resources for demonstrating networking and system implementation.

Constraints

The solution is designed as a basic academic prototype and therefore may not include all features required by a large university. Its performance depends on Internet connectivity, cloud-platform availability, and the resources allocated to the virtual machines. Advanced security, enterprise-scale integration, automated analytics, and high-volume performance testing are outside the primary scope of the project.

Application of Course Knowledge

The project applies cloud computing concepts by mapping the solution to the three major cloud service models.

IaaS: - IaaS is demonstrated through virtual machines created using VMware Workstation Pro, where CPU, memory, storage, operating system, and networking resources are configured.

PaaS: - PaaS is demonstrated through Zoho Creator, which provides the development environment, database, workflows, forms, and application deployment capabilities.

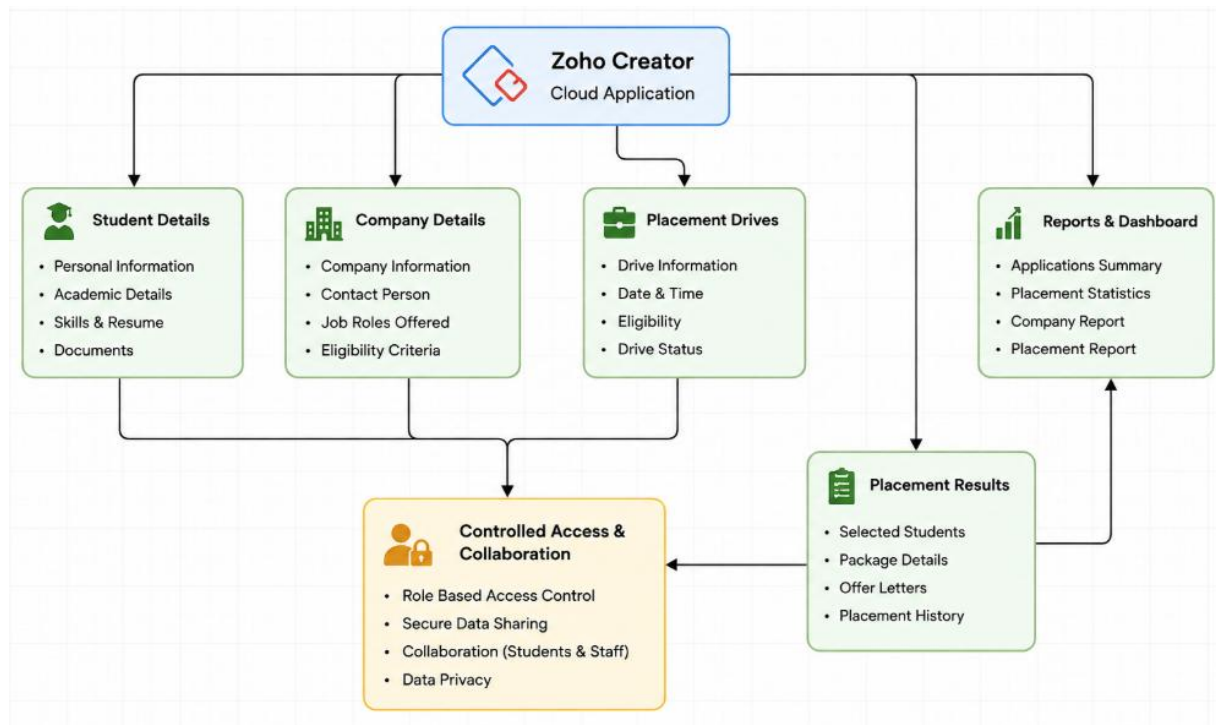
SaaS: - SaaS is demonstrated through the completed cloud-hosted placement application, which users can access through a web browser without installing the application locally.

The project also applies concepts of virtualization, cloud networking, access control, Web APIs, cloud storage, collaboration, and system testing.

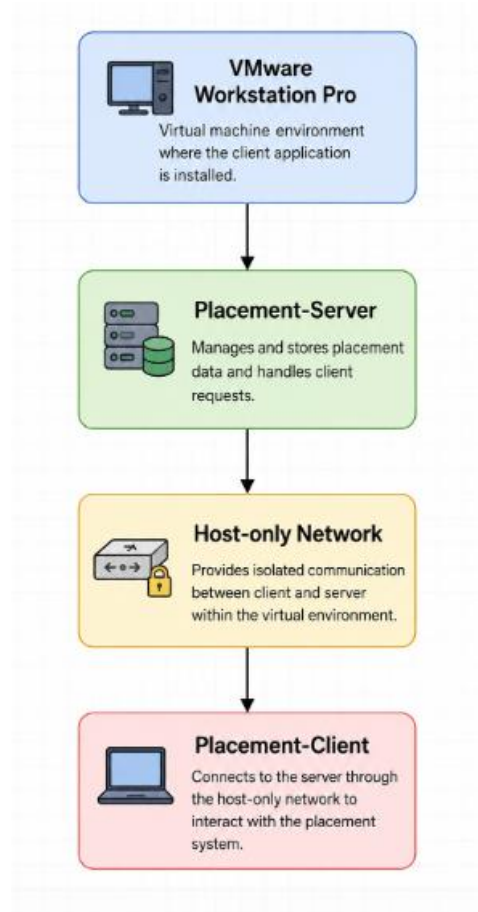
Solution / Design / Methodology

The proposed solution consists of two main parts.

Cloud Application



Virtualization



Both VMs were configured and their communication was tested using the ping command.

Use of Modern Tools

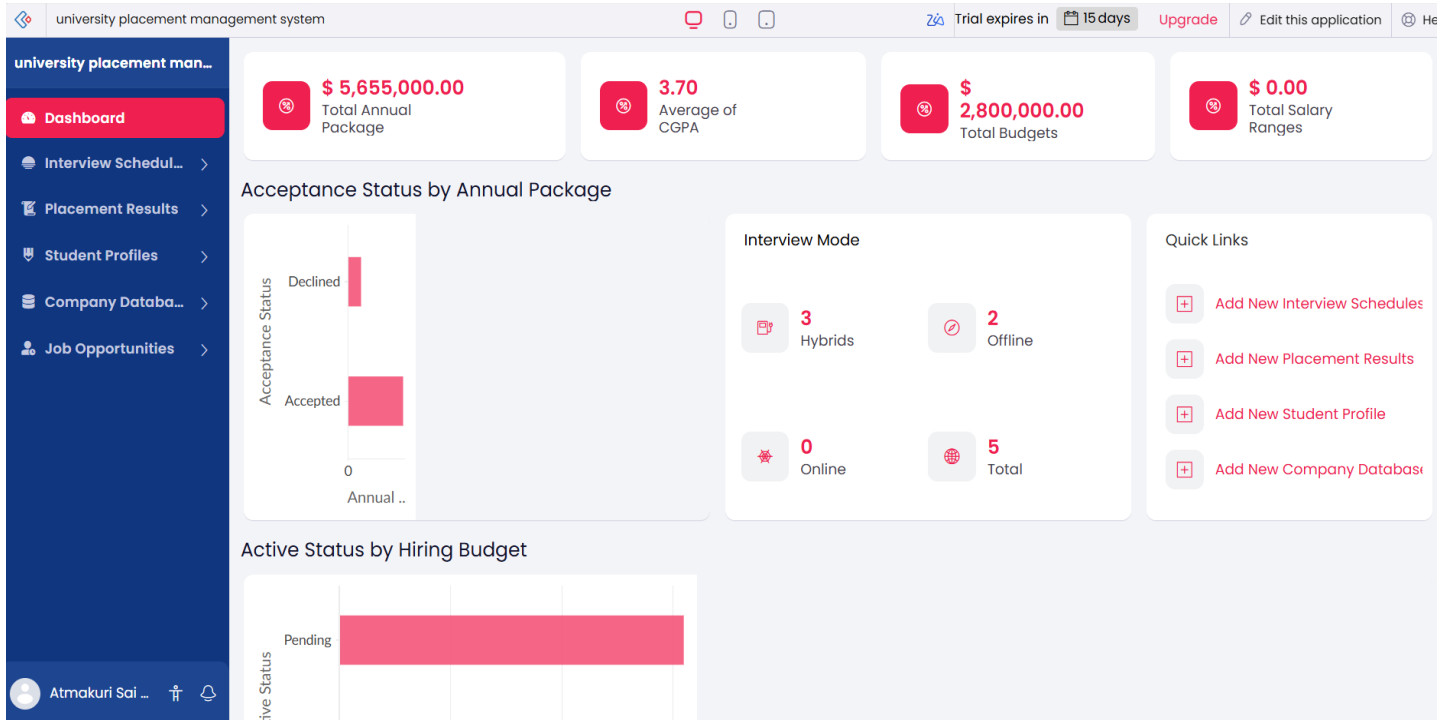
Tool	Purpose
Zoho Creator	Cloud-based placement application
VMware Workstation Pro	Virtual machine creation and configuration
Ubuntu Linux	Operating system for both VMs
Web Browser	Accessing Zoho Creator
Ping	Testing VM network communication

Screenshots of the Zoho application, VM configuration, IP addresses and ping results are provided as evidence.

Results and Validation

The developed system successfully managed placement information through Zoho Creator and provided cloud-based access and collaboration.

1. Zoho Creator placement dashboard.



2. Student Profile module.

university placement management system

Trial expires in 15 days Upgrade Edit this application Help

university placement man...

Dashboard

Interview Schedul... >

Placement Results >

Student Profiles >

+ Student Profile

All Student Profiles

Student Profiles Bo...

Company Databa... >

Job Opportunities >

Atmakuri Sai ...

All Student Profiles

Student Id	Full Name	Email	Phone Number	Date Of Birth	Address	Major	CGPA	Gradu
5	Eva Brown	eva.brown@univ.edu	+19712117100	18-Apr-2004	987 Cedar Ln	Data Science	3.60	
4	David Lee	david.lee@univ.edu	+19724517760	05-Jan-2005	654 Maple Dr	Civil Engineering	3.70	
3	Carol Martinez	carol.martinez@univ.edu	+18506854520	10-Sep-2004	321 Elm St	Business Admin	3.90	
2	Bob Williams	bob.williams@univ.edu	+19707047293	22-Jun-2003	789 Pine Rd	Mechanical Engineering	3.50	
1	Alice Johnson	alice.johnson@univ.edu	+17605581503	15-Mar-2004	456 Oak Ave	Computer Science	3.80	

Showing 5 of 5

3. Company Profile module.

university placement management system

Trial expires in 15 days

Upgrade

Edit this application

Help

university placement man...

Dashboard

Interview Schedul...

Placement Results

Student Profiles

Company Databa...

Company Database

All Company Data...

Active Statuses

Job Opportunities

Atmakuri Sai ...

All Company Databases

Q

+

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☐	Company Name	Industry	Website	Headquarters	Contact Person	Contact Email	Contact Phone
☐	FoodieMart Inc	Retail	www.foodiemart.com	Seattle	James Wilson	james@foodiemart.com	+18138241776
☐	HealthFirst Labs	Biotechnology	www.healthfirst.bio	Boston	Dr. Emily Chen	emily@healthfirst.bio	+18263559017
☐	UrbanBuild Corp	Construction	www.urbanbuild.net	Chicago	David Miller	david@urbanbuild.net	+18074299238
☐	GreenLeaf Energy	Renewable Power	www.greenleaf.org	Austin	Michael Green	mike@greenleaf.org	+18662556110
☐	TechNova Solutions	Software Development	www.technova.com	San Francisco	Sarah Connor	sarah@technova.com	+17659881600

Showing 5 of 5

Show Summary

4. Interview Schedule module.

university placement management system

Trial expires in 15 days

Upgrade

Edit this application

Help

university placement man...

All Interview Schedules

🔍

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☐	Interview Id	Student	Job	Interview Date	Interview Time	Interview Mode	Interviewer Name	Interviewer Email	Interview Status
	5			02-Sep-2026	18:00:00	Hybrid	Pam Beesly	p.beesly@university.edu	Completed
	4			02-Sep-2026	16:45:00	Offline	Jim Halpert	j.halpert@university.edu	Completed
	3			02-Sep-2026	14:15:00	Offline	Dwight Schrute	d.schrute@university.edu	Completed
	2			02-Sep-2026	10:30:00	Hybrid	Michael Scott	m.scott@university.edu	Completed
	1			02-Sep-2026	09:00:00	Hybrid	Sarah Connor	sarah.c@university.edu	Scheduled

Showing 5 of 5

5. Placement Result module.

university placement man...

Dashboard

Interview Schedul... >

Placement Results >

+ Placement Results

All Placement Resu...

Acceptance Status...

Student Profiles >

Company Databa... >

Job Opportunities >

Atmakuri Sai ...

All Placement Results

Placement Id >

Student >

Company >

Job >

Offer Date >

Joining Date >

Annual Package >

Acceptance Status >

Designation >

5

19-Aug-2026

05-Sep-2026

\$ 1,100,000.00

Declined

HR Executive

4

18-Aug-2026

04-Sep-2026

\$ 875,000.00

Accepted

Marketing Specialist

3

17-Aug-2026

03-Sep-2026

\$ 1,450,000.00

Accepted

Product Manag

2

16-Aug-2026

02-Sep-2026

\$ 980,000.00

Accepted

Data Analyst

1

15-Aug-2026

01-Sep-2026

\$ 1,250,000.00

Accepted

Software Engin

Showing 5 of 5

Show Summary >

Two virtual machines were successfully configured:

VM	IP Address
Placement-Server	192.168.101.10
Placement-Client	192.168.101.11

Network Validation

Client → Server:

4 packets transmitted

4 packets received

0% packet loss

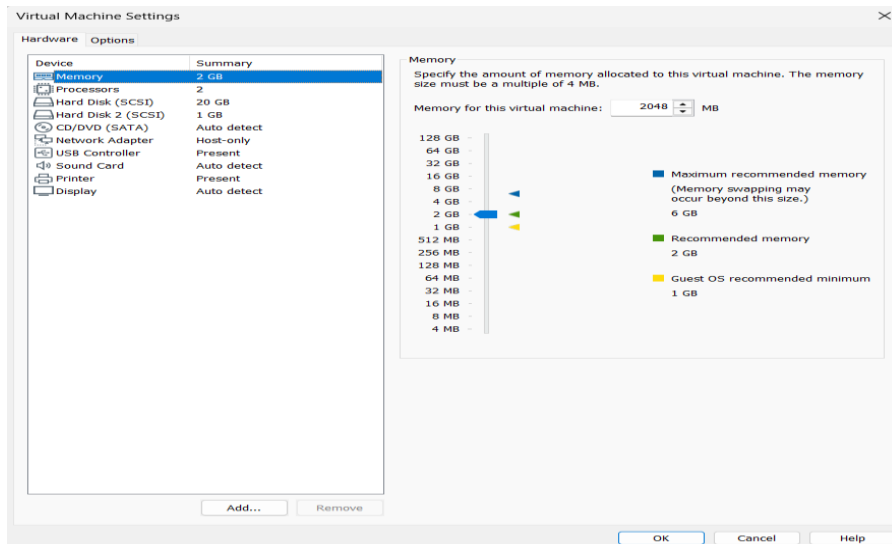
Server → Client:

4 packets transmitted

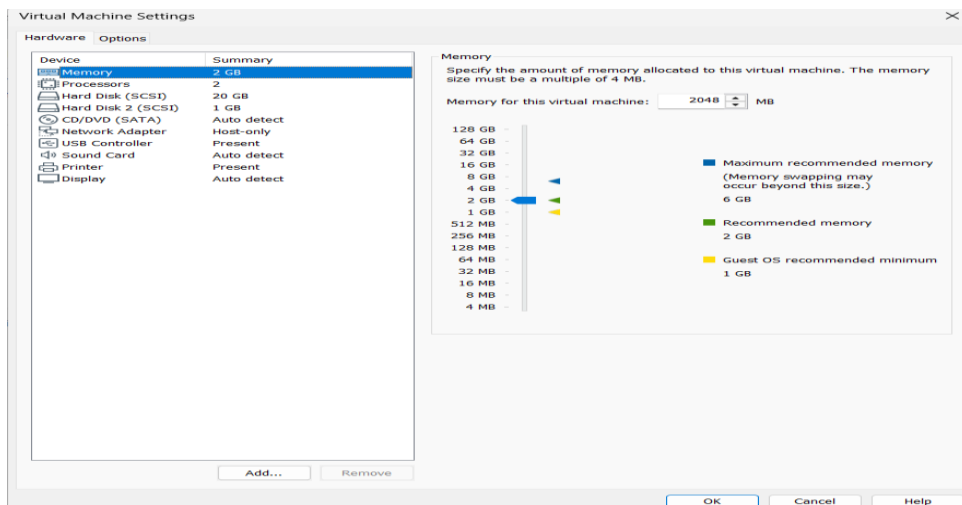
4 packets received

0% packet loss

1. VMware Placement-Server configuration.



2. VMware Placement-Client configuration.



3. Placement-Server IP address.

```
mrudula@~  
File Edit View Search Terminal Help  
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default  
qlen 1000  
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00  
    inet 127.0.0.1/8 scope host lo  
        valid_lft forever preferred_lft forever  
    inet6 ::1/128 scope host  
        valid_lft forever preferred_lft forever  
2: ens33: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group  
default qlen 1000  
    link/ether 00:0c:29:7d:3c:87 brd ff:ff:ff:ff:ff:ff  
    inet 192.168.101.10/24 brd 192.168.101.255 scope global dynamic noprefixroute ens33  
        valid_lft 1747sec preferred_lft 1747sec  
    inet6 fe80::666c:9e7e:7e33:d290/64 scope link noprefixroute  
        valid_lft forever preferred_lft forever
```

4. Placement-Client IP address.

```
mrudula@~  
File Edit View Search Terminal Help  
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default  
qlen 1000  
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00  
    inet 127.0.0.1/8 scope host lo  
        valid_lft forever preferred_lft forever  
    inet6 ::1/128 scope host  
        valid_lft forever preferred_lft forever  
2: ens33: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group  
default qlen 1000  
    link/ether 00:0c:29:5c:07:f3 brd ff:ff:ff:ff:ff:ff  
    inet 192.168.101.10/24 brd 192.168.101.255 scope global dynamic noprefixroute ens33  
        valid_lft 1732sec preferred_lft 1732sec  
    inet6 fe80::4629:c605:110a:572a/64 scope link noprefixroute  
        valid_lft forever preferred_lft forever
```

5. Client-to-Server successful ping.

```
mrudula@~  
File Edit View Search Terminal Help  
mrudula@~$ ping -c 4 192.168.101.10  
PING 192.168.101.10 (192.168.101.10) 56(84) bytes of data:  
64 bytes from 192.168.101.10: icmp_seq=1 ttl=64 time=4.13 ms  
64 bytes from 192.168.101.10: icmp_seq=2 ttl=64 time=1.71 ms  
64 bytes from 192.168.101.10: icmp_seq=3 ttl=64 time=3.06 ms  
64 bytes from 192.168.101.10: icmp_seq=4 ttl=64 time=1.61 ms  
  
--- 192.168.101.10 ping statistics ---  
4 packets transmitted, 4 received, 0% packet loss, time 3007ms  
rtt min/avg/max/mdev = 1.612/2.630/4.134/1.040 ms
```

6. Server-to-Client successful ping.

```
mrudula@~  
File Edit View Search Terminal Help  
mrudula@~$ ping -c 4 192.168.101.10  
PING 192.168.101.10 (192.168.101.10) 56(84) bytes of data:  
64 bytes from 192.168.101.10: icmp_seq=1 ttl=64 time=4.31 ms  
64 bytes from 192.168.101.10: icmp_seq=2 ttl=64 time=1.81 ms  
64 bytes from 192.168.101.10: icmp_seq=3 ttl=64 time=2.24 ms  
64 bytes from 192.168.101.10: icmp_seq=4 ttl=64 time=1.08 ms  
  
--- 192.168.101.10 ping statistics ---  
4 packets transmitted, 4 received, 0% packet loss, time 3005ms  
rtt min/avg/max/mdev = 1.085/2.362/4.312/1.199 ms
```

Therefore, two-way communication between the Placement-Server and Placement-Client was successfully validated. The placement application and virtualization requirements were successfully demonstrated.

Analysis and Engineering Decision

The major engineering decision was to use Zoho Creator for developing the placement application because it provides an integrated cloud environment for creating forms, databases, workflows, reports, and user-access mechanisms without requiring the development and maintenance of a complete server infrastructure.

For virtualization, VMware Workstation Pro was selected to create the Placement-Server and Placement-Client environments. A Host-only network was chosen because it allows controlled communication between the virtual machines while keeping the demonstration environment isolated from the external network. This configuration is suitable for testing network communication and cloud-related implementation concepts in an academic environment.

Advantages

- Centralized placement information management.
- Easy access through a web browser.
- Role-based access for students and placement staff.
- Reduced requirement for physical infrastructure.
- Easy collaboration through shared cloud resources.

Limitations

- The system depends on Internet and cloud-platform availability.
- Zoho Creator features may vary according to the subscription plan.
- The virtual machines are limited by the host computer's hardware.
- The prototype may not represent the performance requirements of a large production university system.
- Advanced analytics and intelligent placement recommendations are not included.

Engineering Decision

The final design combines Zoho Creator, VMware Workstation Pro, and controlled networking to create a simple but practical cloud placement solution. Zoho Creator was selected for application development and cloud data management, while VMware Workstation Pro provides the virtualized infrastructure environment for implementation and testing. The design provides a balance between simplicity, security, accessibility, cost, and ease of implementation.

Pseudo Code

START

1. Open the Zoho Creator Placement Management application.
2. Add student details.
3. Add company details.
4. Create placement drive details.

5. Store and manage placement results.
 6. Provide controlled access to students and placement staff.
 7. Allow users to access and share placement information through the cloud.
 8. Create Placement-Server virtual machine in VMware.
 9. Create Placement-Client virtual machine in VMware.
 10. Configure CPU, RAM, storage and network for both VMs.
 11. Assign Host-only network to both VMs.
 12. Obtain the IP address of both VMs.
 13. Test Client → Server communication using ping.
 14. Test Server → Client communication using ping.
 15. If communication is successful,
 Record the result.
 Else,
 Check the network configuration.
 16. Verify that all placement and virtualization requirements are satisfied.
- END

Broader Considerations

The proposed system can be expanded to support a larger university environment. Future versions could integrate email or notification services, automated eligibility checking, analytics dashboards, resume processing, company APIs, mobile access, and intelligent job recommendations. Stronger authentication and auditing mechanisms can also be introduced for handling sensitive institutional information. The architecture can additionally be integrated with existing university student information systems to avoid duplicate data entry.

Conclusion

The project successfully demonstrates the design and implementation of a cloud-based university placement data management and access solution using Zoho Creator. The solution provides centralized management of student and placement information with controlled access for different users. Zoho Creator demonstrates the PaaS and SaaS aspects of cloud computing, while VMware Workstation Pro demonstrates virtualization and infrastructure concepts. The use of a Host-only network provides a controlled environment for communication and testing. Overall, the system demonstrates how cloud computing, virtualization, networking, security, and collaboration can be combined to develop an efficient placement management solution.

Possible Improvements

- Implement multi-factor authentication for staff and administrators.
- Add automated email/SMS notifications for interviews and selection results.
- Develop a mobile-friendly interface or mobile application.
- Add AI-based job recommendation and student-job matching.
- Integrate the application with the university's existing student database.

Student Reflection

I was responsible for the complete implementation and practical work of the project. I developed and configured the placement management application in Zoho Creator and implemented the required placement data management, access control, and collaboration features. I also created and configured the Placement-Server and Placement-Client virtual machines using VMware Workstation Pro. I configured their network settings and established communication between both virtual machines using a Host-only network. Finally, I tested the complete implementation, verified the results, identified and resolved configuration issues, and prepared the required documentation and screenshots.

Modules/Tasks Completed

- Designed and developed the Cloud-Based Placement Data Management System using Zoho Creator.
- Created and configured modules for Student Management, Company Management, Job Opportunities, Applications, Interviews, and Placement Records.
- Implemented user authentication and role-based access control for students and placement staff.
- Configured forms, workflows, reports, and dashboards in Zoho Creator.
- Implemented placement data storage and controlled access to shared cloud resources.
- Created and configured Placement-Server and Placement-Client virtual machines using VMware Workstation Pro.
- Configured a Host-only network and established communication between the virtual machines.
- Tested the cloud application, VM connectivity, access permissions, and application workflows.
- Verified the overall system functionality and prepared screenshots, test results, and project documentation.

Brief Description of Work

I developed and configured a cloud-based university placement management system using Zoho Creator. I created the required forms, workflows, reports, and access permissions for managing student, company, job, application, interview, and placement data. I also configured Placement-Server and Placement-Client virtual machines using VMware Workstation Pro, established communication using a Host-only network, tested the complete system, and documented the implementation with screenshots and results.

Outcome/Learning

Through this project, I gained practical knowledge of cloud computing, Zoho Creator, IaaS, PaaS, SaaS, virtualization, VMware Workstation Pro, Ubuntu, networking, access control, cloud collaboration, and system testing. I learned how to develop and deploy a cloud-based application, configure virtual machines and networks, manage controlled user access, troubleshoot implementation issues, and validate the complete system.

References

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